Craig M. Nielsen

Middleton, WI | 608-504-7635 | nielsen.craigm@gmail.com

EDUCATION

New York University | College of Arts and Science | New York, NY Sep 2023-May 2027
Bachelor of Arts | Data Science | Dean's Presidential Honors | GPA 3.95
Minors: Mathematics, Music

- **Relevant Coursework:** Honors Real Analysis I, Linear Algebra and Optimization for Machine Learning, Fundamentals of Machine Learning, Calculus III, Probability & Statistics, Causal Inference, Data Management and Analysis, Intermediate Microeconomics, Linear Algebra
- **Involvement:** NYU All-University Jazz Orchestra Bassist, Music Mentors Collaborative Instructor, NYU Orchestra Bassist, NYU Prague Blog

HONORS AND AWARDS

Awarded $1250 from NYU Dean's Undergraduate Research Fund for research on *Why People Listen to Sad Music*.
Applying for $1250 from Dean's Undergraduate Research Fund for research on *Mapping Timbre Space*.
Awarded a fully funded study abroad opportunity through NYU's Dean's Presidential Honors Society

RESEARCH EXPERIENCE

Research Assistant | *Fox Lab NYU* | New York, NY Jan 2026-Present

- Meet weekly with Dr. Pascal Wallisch, Department of Data Science, to discuss research and analyze data
- Running participants weekly for a study on pattern recognition in chess
- Awarded a $1250 grant on Why People Listen to Sad Music

Research Assistant | *Dr. Laurence Maloney's Lab NYU* | New York, NY July 2026-Present

- Meet weekly with Dr. Laurence Maloney, Department of Psychology, to discuss experiment design and grant writing
- Designing a study investigating the geometry of timbre perception
- Applying for a $1250 grant on Mapping Timbre Space

Research Assistant | *CROWE* University of Wisconsin-Madison* | Madison, WI June-July 2025

- Worked with Dr. Ananth Seshadri, Department of Economics, and met with a post-doc researcher weekly
- Created a model measuring the causal effect of a tax credit on manufacturing employment
- Created data visualizations depicting said tax credit's effect
- Contributed to the publication of this report

RESEARCH INTERESTS

Out-of-distribution generalization, high-dimensional statistics, machine learning for the social sciences.

PROJECTS

Michelin Star Restaurant App | *NYU* | New York, NY Jan-May 2026

- Developed in Fundamentals of Machine Learning course with Kyunghyun Cho, PhD.
- I am entirely responsible for the recommendation engine of the app.
- Domain-shifted large Yelp data to sparse Michelin star restaurant reviews in NYC using the Python library Hugging Face. The model estimated rating distributions with a multi-headed Laplace distribution, using a corruption function to focus the model on target restaurants.

Effect of Corporate Income Tax on Manufacturing Employment June-July 2025
CROWE University of Wisconsin-Madison* | Madison, WI

- Created a difference-in-differences model on the effect of corporate income tax on manufacturing employment in Wisconsin and bordering states
- Received and aggregated data from the Quarterly Census of Employment and Wages
- Cleaned and analyzed data using Pandas and Matplotlib in Python

*CROWE - Center for Research on the Wisconsin Economy

ServiceNow AI | *Spectrum Brands* | Middleton, WI May-Aug 2024

- Created an analysis of business costs and benefits for implementing ServiceNow AI at a Fortune 500 company
- Presented to company leaders, including the CFO, with a team of 3 other interns

TEACHING EXPERIENCE

Instructor | *Lavner Education* | New York, NY June-Aug 2026

- Planned and taught introductory computer science and machine learning courses
- Courses included *Agentic AI and Machine Learning* and *Python & App Development*
- Taught private lessons

OTHER EXPERIENCE

Desktop Support Intern | *Spectrum Brands* | Middleton, WI May-Aug 2024

- Duties included responding to service desk calls, resolving ServiceNow tickets through ITIL best practices, and customer service
- Learned SAP basics through assisting users with SAP-related tickets and meetings with company IT experts
- Used Azure Device Management to enforce security protocols and assist users

Customer Service | *Cafe Zupas and Prairie Cafe* | Middleton, WI June 2021-Aug 2023

SKILLS

Programming Languages: Java, JavaScript, Python, SQL, R, MATLAB
Python Packages: PyTorch, TensorFlow, Matplotlib, Pandas, NumPy, scikit-learn
Business Applications: Power BI, Excel, ServiceNow, MS Office Suite
Languages: German, intermediate B2 proficiency

PAID MUSICAL PERFORMANCES

La Brioche | Madison, WI July 2024-25
Harmony Bar and Grill | Madison, WI July 2024
Middleton High School Band & Orchestra Parents Association | Middleton, WI May 2023
Dane County Buy Local | Middleton, WI Feb 2023

REFERENCES

Pascal Wallisch, PhD: Data Science, Neural Science, and Psychology | New York University

Ananth Seshadri, PhD: Economics | University of Wisconsin - Madison

Laurence Maloney, PhD: Psychology, Neural Science | New York University